

Bayesian Bandit Encoding: Analysis on a Two-State Markov Channel

Extended version with proofs

Bhaskar Krishnamachari

Ming Hsieh Department of Electrical and Computer Engineering

University of Southern California, Los Angeles, CA, USA

bkrishna@usc.edu

Abstract

We introduce Bayesian bandit encoding (BBE), a framework that selects how to encode packets for a channel whose parameters are unknown. BBE has two components. A Bayesian learner maintains posteriors over the quantities that determine performance, updated from the noise patterns recovered by a guessing random additive noise decoding (GRAND) receiver and from packet acknowledgments. The transmitter selects an encoding strategy by Thompson sampling from these posteriors, and the receiver uses the learned channel parameters to order its noise guesses. We analyze BBE on a two-state Markov burst channel, where the transmitter decides whether to interleave a random linear code. For this channel, the GRAND query order has a closed form, the block error rate has a closed-form expression under an independent false-hit model, and interleaving to depth D shrinks the channel's memory from λ to λ^D . We show that with a matched decoder, interleaving is asymptotically dominated for any positive delay penalty. When every frame reveals its full noise sequence, BBE's regret is bounded uniformly in time under a margin condition, with explicit constants. In practice GRAND reveals the noise only of codewords it decodes, so the encoding choice determines what the receiver learns. Simulations show that the BBE variants whose selection also uses acknowledgments interleave early when the initial belief is that the channel is memoryless, learn the channel memory faster, and achieve lower regret than selection from noise patterns alone. This extended version contains the proofs and additional numerical results, and is available at <https://github.com/anrgusc/bbe-2smc>.

1 Introduction

Encoders and decoders are usually designed for an assumed channel. Interleaving is a standard example. A block interleaver spreads each burst of errors over many codewords, so that a decoder designed for independent errors can correct them, and it pays for this with delay. A decoder that knows the burst statistics can exploit the bursts instead, and for such a decoder the interleaver discards useful structure [3]. The better encoding therefore depends on the channel and on how much the receiver knows about it, and neither is known when a link starts.

We introduce Bayesian bandit encoding (BBE), which learns the channel while the link operates and uses what it learns at both ends. A Bayesian learner maintains a posterior over the quantities that determine performance. Its main evidence comes from the receiver, which runs guessing random additive noise decoding (GRAND) [4]. GRAND guesses noise patterns in decreasing order

of likelihood and stops at the first one that turns the received word into a codeword, so every decoded codeword also reveals the noise the channel applied. Packet acknowledgments provide further evidence. The transmitter selects the encoding of each frame by Thompson sampling from the posterior, and the receiver orders its guesses by the learned channel parameters. Encoder and decoder are adapted from the channel and action posteriors in a single loop. Two unpublished manuscripts demonstrated BBE in simulation, on a channel with several on-off interferers [5] and with catalogs of more than a thousand code and decoder configurations [6]. This paper gives the first analysis.

We study BBE on a two-state Markov burst channel, where the transmitter decides in each frame whether to send a random linear code (RLC) plain or through a block interleaver. On this channel the GRAND query order and the posterior update have closed forms, the block error rate (BLER) has a closed-form expression under an independent false-hit model, and the instance exhibits the effect that motivates BBE: the best encoding changes as the receiver learns. At our operating point, interleaving lowers the BLER of a memoryless decoder from 0.089 to 0.039. Once the decoder has learned the bursts, plain transmission reaches 0.0097, nine times lower than at the start and three times lower than interleaving. BBE finds this switch on its own. Our contributions are:

- *The BBE framework.* Bayesian learning from decoded noise and acknowledgments, Thompson-sampling selection of the encoding, and a GRAND query order driven by the learned channel (Section 2), with the explicit update and sampling rules for the two-state channel (Section 4).
- *Analysis of GRAND on the two-state channel.* The 2^n noise patterns fall into $O(n^2)$ classes of equally likely patterns, which gives the query order and a closed-form BLER under an independent false-hit model, validated against an explicit RLC decoder, and an interleaved codeword sees a two-state channel with memory λ^D (Section 3).
- *Interleaving versus decoder knowledge.* Interleaving raises the entropy rate of each codeword's noise, so with a matched decoder it is asymptotically dominated for any positive delay penalty, while with a memoryless decoder it lowers the variance of the block weight and can win at practical lengths (Section 5).
- *Learning under full feedback.* When every frame reveals its noise, the decoder reaches the matched query order after finitely many frames, its loss near that order is bounded, and the expected number of frames more than δ from optimal is bounded uniformly in time, with explicit constants in the channel, the query budget and the penalty (Sections 6 and 8).
- *Learning under decoded feedback.* The bias of the channel estimate shrinks with the BLER of the decoder that produced the data, and at the operating point of the simulations the long-run estimate gives a decoder whose BLER equals the matched one to four digits (Section 7).
- *Closed-loop simulations.* All BBE variants learn the bursts and have a fraction of the regret of a conventional interleaved link. BBE-ACK and BBE-hybrid, whose selection uses acknowledgments, interleave early and move to plain transmission within the first tens of frames when the prior favors a memoryless channel (Section 9).

Related work. Three lines of work are related. The first is GRAND itself. Its soft-information variants [7, 8] use reliability information to order queries, and for channels with memory a GRAND decoder with a known Markov model outperforms interleaving [3]. Universal noise-guessing decoders estimate an unknown finite-state channel within the decoding rule [9]. In BBE the

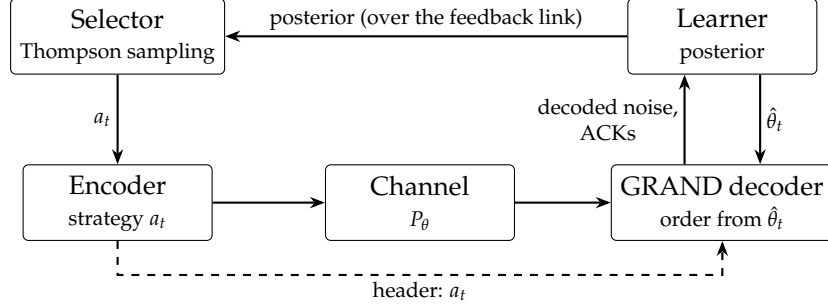


Figure 1: One frame of BBE. The posterior learned from decoded noise and acknowledgments selects the encoding and sets the GRAND query order.

Algorithm 1 One frame of Bayesian bandit encoding

- 1: The receiver sets its GRAND order from the posterior mean $\hat{\theta}_t$
 - 2: The transmitter draws a sample from the posterior, plays the action a_t with the lowest sampled cost, and announces a_t in a header
 - 3: GRAND decodes each codeword in the order implied by $\hat{\theta}_t$ and a_t
 - 4: Decoded noise updates the channel posterior; ACKs update the per-action posteriors
-

learned channel also drives the transmitter’s choice of encoding. The second line relates guessing to entropy [10, 11, 12], and we use the entropy rate to compare interleaved and plain transmission. The third is bandit learning with structure. Bandit algorithms have been applied to rate adaptation [1, 2], and when all arms depend on a common parameter that is observed whichever arm is played, regret can be bounded in time [13, 14]. In BBE the channel is that shared parameter, and [Theorem 1](#) gives the constants. Under decoded feedback the observations depend on the arm, as in partial monitoring [17] and bandits with graph-structured feedback [18]. BBE’s acknowledgment-based variants are one way to account for this, and information-directed sampling [19] is another.

2 Bayesian Bandit Encoding

The channel belongs to a parametric family $\{P_\theta : \theta \in \Theta\}$. In each frame the transmitter selects an action a from a finite set $\mathcal{A}$. An action fixes the encoding strategy (code, rate, interleaving) and may also fix decoder parameters such as the GRAND query budget. A *performance model* gives the expected cost $J_a(\theta; \hat{\theta})$ of action a when the channel is θ and the receiver orders its queries using the noise model $P_{\hat{\theta}}$. [Figure 1](#) and [Algorithm 1](#) describe one frame.

The learner holds two kinds of posterior. The *channel posterior* over θ is updated from decoded noise, and its mean $\hat{\theta}_t$ configures the decoder. The *action posteriors* are Beta distributions on each action’s codeword loss rate, updated from acknowledgments. The selector can draw from either or both, which gives three instances of BBE.

- *BBE-noise* draws $\tilde{\theta}_t$ from the channel posterior and plays $a_t = \arg \min_a J_a(\tilde{\theta}_t; \hat{\theta}_t)$. The first argument is a channel drawn from the posterior, and the second is the belief that configures the decoder in the coming frame, so while the posterior is broad the prediction includes the cost of decoding with a mismatched noise model. Every action is scored through one channel model, so an observation made under one action updates the predicted cost of all of them.

Table 1: Notation.

$\theta = (p, q)$	transition probabilities $\mathbb{P}(1 0)$ and $\mathbb{P}(1 1)$ of the noise chain
$\pi = (\pi_0, \pi_1), \lambda$	stationary law, error rate $\pi_1 = p/(1 - q + p)$, memory $\lambda = q - p$
$H(\theta)$	entropy rate $\pi_0 h(p) + \pi_1 h(q)$
n, k, r	code length, dimension, redundancy $r = n - k$
D, M	interleaver depth (codewords per frame), channel uses per frame $M = Dn$
A, B	plain and interleaved transmission
$\theta^{(D)}$	parameters of the chain seen by one interleaved codeword
Q, ε	GRAND query budget per codeword, false-hit probability 2^{-r}
c, c^*	delay penalty for interleaving, critical penalty
$J_a(\theta; \hat{\theta})$	expected cost of action a on channel θ with decoder belief $\hat{\theta}$
$J^*(\theta)$	optimal cost $\min_a J_a(\theta; \theta)$
$\hat{\theta}_t, \tilde{\theta}_t$	posterior mean and posterior sample in frame t
$\hat{\pi}_1, \kappa$	initial error-rate estimate, prior strength (pseudo-transitions per parameter)
w, m, s, e	pattern weight, number of bursts, first bit, last bit
β_b, γ_b	cost of a new burst and of a further error within a burst
ρ_0, ρ_1, δ	margin radii and tolerance of Assumption 1
R_T	regret over T frames

- *BBE-ACK* draws a loss rate for each action from its action posterior and adds the delay penalty. It measures each action’s performance directly.
- *BBE-hybrid* uses the action posteriors with the channel model’s prediction $\text{BLER}_a(\tilde{\theta}_t; \hat{\theta}_t)$ added as prior pseudo-counts.

In all three, the decoder is configured from the channel posterior, and [Section 4](#) gives the explicit rules for the two-state channel.

BBE requires a posterior update from decoded noise and a performance model. For the multi-interferer channel of [\[5\]](#) the posterior was a point estimate with a confidence score, and in [\[6\]](#) the performance model was learned offline from simulated decoder statistics. On the two-state channel the update is conjugate and the performance model has a closed form, which is what makes the analysis below possible; [Sections 3](#) and [4](#) state where each is approximate. [Table 1](#) collects the notation.

3 The Two-State Markov Instance

3.1 Channel, code and strategies

The channel flips bit t when $Z_t = 1$, where $Z_1, Z_2, \dots$ is a stationary two-state Markov chain with transition matrix

$$P = \begin{pmatrix} 1-p & p \\ 1-q & q \end{pmatrix}, \quad p = \mathbb{P}(Z_{t+1} = 1 | Z_t = 0), \quad q = \mathbb{P}(Z_{t+1} = 1 | Z_t = 1). \quad (1)$$

We write $\theta = (p, q) \in (0, 1)^2$ and denote the stationary law by $\pi = (\pi_0, \pi_1)$, with error rate $\pi_1 = p/(1 - q + p)$. The memory $\lambda = q - p$ is the second eigenvalue of P , and $H(\theta) = \pi_0 h(p) + \pi_1 h(q)$ is the entropy rate in bits, where h is the binary entropy function. Bursts last $1/(1 - q)$ bits on average, and $p = q$ gives a memoryless channel.

A frame carries D codewords of a binary (n, k) random linear code (RLC) with redundancy $r = n - k$, a total of $M = Dn$ channel uses. Strategy A sends the codewords one after another. Strategy B writes them as the rows of a $D \times n$ array and sends the array column by column, so codeword j occupies channel uses $j, j + D, j + 2D, \dots$ and cannot be decoded until the frame has arrived. The cost of a frame is the fraction of codewords lost plus a penalty $c \geq 0$ for the delay of interleaving,

$$J_a(\theta; \hat{\theta}) = \text{BLER}_a(\theta; \hat{\theta}) + c \mathbf{1}\{a = \text{B}\}, \quad J^*(\theta) = \min_a J_a(\theta; \theta). \quad (2)$$

Lemma 1. *The noise seen by one codeword under B, $(Z_j, Z_{j+D}, Z_{j+2D}, \dots)$, is a stationary two-state Markov chain with stationary law π and memory λ^D , that is, with parameters $\theta^{(D)} = (\pi_1(1 - \lambda^D), \pi_1 + \pi_0\lambda^D)$.*

Proof. Write $\Pi = \mathbf{1}\pi^\top$. Since $\pi^\top P = \pi^\top$ and $P\mathbf{1} = \mathbf{1}$, we have $P\Pi = \Pi P = \Pi$ and $\Pi^2 = \Pi$, and one checks $P = \Pi + \lambda(I - \Pi)$ entrywise. The matrices Π and $I - \Pi$ are complementary projections, so $P^D = \Pi + \lambda^D(I - \Pi)$. The subsampled process is Markov with transition matrix P^D and is stationary because $\pi^\top P^D = \pi^\top$. Its entries $(P^D)_{01} = \pi_1(1 - \lambda^D)$ and $(P^D)_{11} = \pi_1 + \pi_0\lambda^D$ give the claim. $\square$

The receiver decodes strategy B by applying the same map to its belief, $\hat{\theta} \mapsto \hat{\theta}^{(D)}$. At the operating point used later ($\pi_1 = 0.01$, mean burst length 4, so $p \approx 0.0025$, $q = 0.75$ and $\lambda \approx 0.75$) and with $D = 8$, an interleaved codeword sees memory $\lambda^8 \approx 0.10$.

3.2 GRAND on the two-state channel

GRAND with abandonment [4] queries noise patterns $z^{(1)}, z^{(2)}, \dots$ in decreasing order of probability under its belief and returns the first one for which $y \oplus z^{(i)}$ is a codeword. It declares an erasure after Q queries. It errs if the true pattern lies beyond the budget or if an earlier pattern passes the parity check. For an RLC such a false hit occurs with probability 2^{-r} per query.

For a pattern $z \in \{0, 1\}^n$ let w be its weight, m its number of bursts (maximal runs of ones), $s = z_1$ and $e = z_n$.

Lemma 2. *Under θ ,*

$$\mathbb{P}_\theta(Z = z) = \pi_s(1 - p)^{n_{00}} p^{n_{01}} (1 - q)^{n_{10}} q^{n_{11}}, \quad (3)$$

with $n_{01} = m - s$, $n_{10} = m - e$, $n_{11} = w - m$ and $n_{00} = n - 1 - w - m + s + e$. The probability depends on z only through (w, m, s, e) . The all-zero and all-one patterns form singleton classes. For every other key with $m \geq 1$ the class has $\binom{w-1}{m-1} \binom{n-w-1}{G-1}$ members, where $G = m - 1 + (1 - s) + (1 - e)$.

Proof. Each burst that does not start at position 1 is entered by a $0 \rightarrow 1$ transition, and each burst that does not end at position n is left by a $1 \rightarrow 0$ transition. The remaining $w - m$ transitions inside bursts are $1 \rightarrow 1$, and the rest are $0 \rightarrow 0$. A pattern in the class is determined by a composition of w into m positive burst lengths and a composition of the $n - w$ zeros into the G gaps that must be non-empty, namely the $m - 1$ interior gaps, a leading gap if $s = 0$ and a trailing gap if $e = 0$. $\square$

A GRAND query order is thus a sort of the $O(n^2)$ classes. Classes that tie under the belief form a *tie group*, and we query the members of a tie group in uniformly random order. Some ties hold for every θ . By reversibility, $\pi_1(1 - q) = \pi_0 p$, so the classes $(w, m, 1, 0)$ and $(w, m, 0, 1)$ always tie.

For patterns with $s = e = 0$ the lemma gives $-\log \mathbb{P}_\theta(z) = \text{const} + m\beta_b + (w - m)\gamma_b$ with

$$\beta_b = \log \frac{(1 - p)^2}{p(1 - q)}, \quad \gamma_b = \log \frac{1 - p}{q}. \quad (4)$$

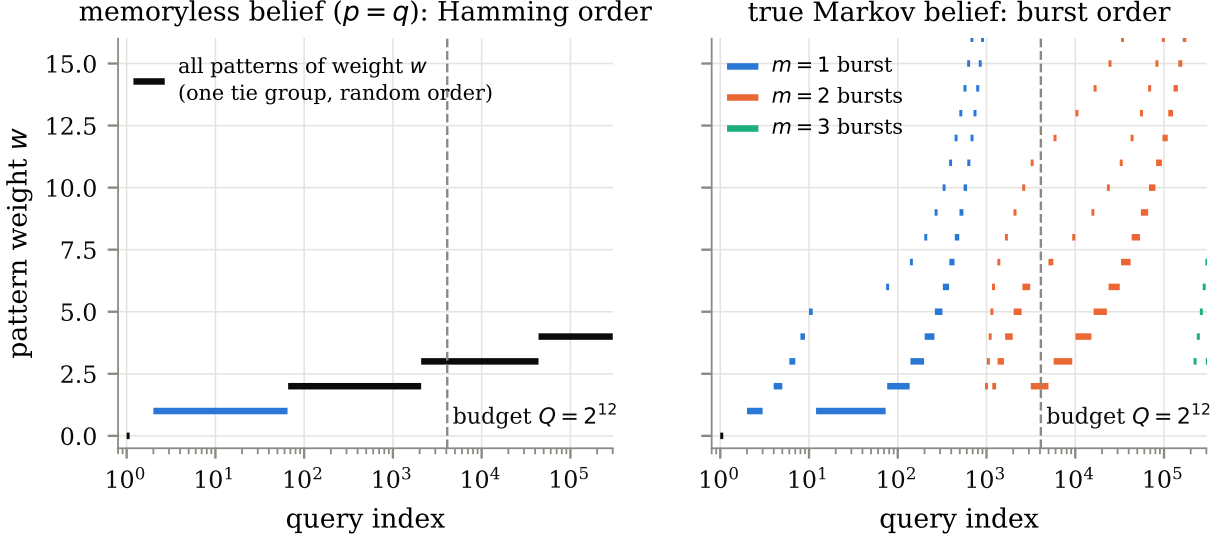


Figure 2: GRAND query order for $n = 64$ under a memoryless belief (left) and under the true chain with $\pi_1 = 0.01$ and mean burst length 4 (right). Each segment is a tie group, placed at its weight over the query indices it occupies and colored by its number of bursts m when that number is unique. The dashed line is the budget $Q = 2^{12}$.

Each new burst adds β_b to the cost of a pattern, and each further error in a burst adds γ_b . The difference $\beta_b - \gamma_b = \log \frac{q(1-p)}{p(1-q)}$ is positive exactly when $\lambda > 0$. When $p = q$ the two costs coincide and the order reduces to the Hamming-weight order of classical GRAND. At the operating point $\beta_b \approx 7.4$ and $\gamma_b \approx 0.29$, so one additional burst costs as much as about 26 additional errors within a burst. Figure 2 shows the first queries under each belief. Within $Q = 2^{12}$ queries the matched decoder covers every single burst of up to 31 errors, whereas the Hamming decoder covers the 2,081 patterns of weight at most two and about 5% of the 41,664 patterns of weight three.

Proposition 1. Sort the tie groups of $\hat{\theta}$ by decreasing likelihood, let A_g be the number of patterns in groups ranked before group g , and let $\varepsilon = 2^{-r}$. Under the random-codebook approximation, in which each incorrect query passes the parity check independently with probability ε ,

$$1 - \text{BLER}(\theta; \hat{\theta}) = \sum_g \frac{\mathbb{P}_\theta(Z \in g)}{|g|} \sum_{j=1}^{L_g} (1 - \varepsilon)^{A_g + j - 1}, \quad L_g = \min\{|g|, (Q - A_g)^+\}. \quad (5)$$

Proof. Because queries within each tie group are uniformly randomly ordered, every pattern in g has a uniform rank within that group. Given $Z \in g$, the rank of Z is therefore $A_g + U$ with U uniform on $\{1, \dots, |g|\}$. Decoding succeeds iff $A_g + U \leq Q$ and none of the $A_g + U - 1$ earlier queries is a false hit, which has probability $(1 - \varepsilon)^{A_g + U - 1}$. Averaging over U and g gives (5). $\square$

The proof also gives a decoding simulator that needs no codebook. It draws Z , then its rank, then the position of the first false hit as a geometric variable, and compares the two. The simulations use this simulator, and (5) is the performance model in Algorithm 1. Table 2 and Fig. 3 compare (5) with a real GRAND decoder on an explicit RLC(64, 44). The differences lie within Monte Carlo error.

Table 2: BLER from (5) (model) and from a real GRAND decoder on one explicit RLC(64,44) with 2×10^4 blocks per entry (real), at the operating point. The standard error of the real values is at most 0.0035.

strategy, decoder belief	$Q = 2^4$		$Q = 2^7$		$Q = 2^{10}$		$Q = 2^{12}$	
	model	real	model	real	model	real	model	real
A, matched	.140	.136	.087	.086	.0118	.0130	.0097	.0101
A, memoryless	.147	.145	.117	.122	.105	.104	.0887	.0902
B, matched	.371	.367	.115	.115	.0795	.0795	.0292	.0299
B, memoryless	.372	.376	.143	.145	.0958	.0949	.0392	.0415

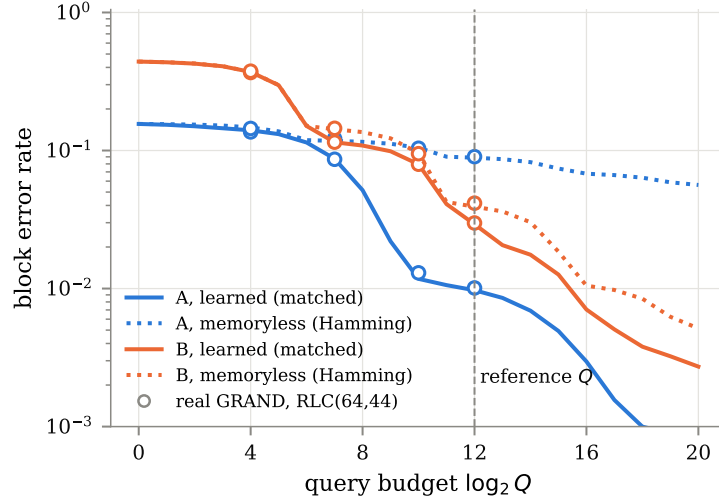


Figure 3: BLER versus query budget for both strategies and both decoder beliefs (RLC(64,44), $D = 8$). Lines are computed from (5); markers are Monte Carlo estimates for a real GRAND decoder.

4 BBE on the Two-State Channel

This section spells out each step of [Algorithm 1](#) for the two-state channel: what a frame reveals, how the posterior is updated, how the decoder is set, and how each variant selects the encoding. [Algorithm 2](#) collects the steps.

4.1 What a frame reveals

When GRAND decodes a codeword correctly, the pattern it returns is the noise on that codeword's channel uses. An earlier false hit, which occurs with probability at most $Q2^{-r}$ per codeword, returns a wrong pattern. Acknowledgments count such undetected errors as losses, as an outer error-detecting code would. In our simulations they occur for about 2×10^{-4} of the codewords and the channel update uses the returned pattern; with an outer code they can be excluded from the update as well. The receiver writes the decoded patterns back into channel order and marks the bits of lost codewords as unknown. It then counts, over adjacent channel uses $(t, t+1)$ whose bits are both known, the transitions

$$n_{ij} = \#\{t : Z_t = i, Z_{t+1} = j\}, \quad i, j \in \{0, 1\}. \quad (6)$$

For example, the known segment 0011100 gives $n_{00} = 2$, $n_{01} = 1$, $n_{11} = 2$ and $n_{10} = 1$. Under strategy A a decoded codeword covers n consecutive channel uses and contributes $n - 1$ transitions, plus one more if the next codeword is also decoded. Under B adjacent channel uses belong to consecutive codewords, so a transition is known when both codewords are decoded, and a fully decoded frame yields all $M - 1$ transitions.

We distinguish two feedback models. Under *decoded feedback*, the model above, the counts come from decoded codewords only. Under *full feedback* the learner sees the noise sequence of every frame, as if every codeword were decoded. Full feedback is the setting of [Section 6](#), and decoded feedback is the setting of [Section 7](#) and of the simulations.

The two parameters are learned at very different speeds. Transitions out of state 0 occur about $\pi_0 M$ times per frame and transitions out of state 1 about $\pi_1 M$ times. Errors are rare, so $\pi_1 \ll \pi_0$, and the burst persistence q is learned about π_0/π_1 times more slowly than p .

4.2 Bayesian update

Lemma 3. *Let $p \sim \text{Beta}(\alpha_0, \beta_0)$ and $q \sim \text{Beta}(\alpha_1, \beta_1)$ be independent. Conditional on the first bit of each run of consecutive known noise bits, the posterior after observing transition counts $(n_{00}, n_{01}, n_{10}, n_{11})$ is $p \sim \text{Beta}(\alpha_0 + n_{01}, \beta_0 + n_{00})$ and $q \sim \text{Beta}(\alpha_1 + n_{11}, \beta_1 + n_{10})$, again independent.*

Proof. Given its first bit, the likelihood of a run is $p^{n_{01}}(1-p)^{n_{00}} \cdot q^{n_{11}}(1-q)^{n_{10}}$, a product of two Bernoulli likelihoods, one for each current state. Each factor is conjugate to its Beta prior. $\square$

The update after a frame is therefore

$$\alpha_0 += n_{01}, \quad \beta_0 += n_{00}, \quad \alpha_1 += n_{11}, \quad \beta_1 += n_{10}, \quad (7)$$

and the posterior mean is $\hat{\theta} = (\hat{p}, \hat{q})$ with $\hat{p} = \alpha_0/(\alpha_0 + \beta_0)$ and $\hat{q} = \alpha_1/(\alpha_1 + \beta_1)$. The whole posterior is four numbers, and the receiver can return it to the transmitter in a few bytes per frame. Under full feedback, [Lemma 3](#) makes this the exact posterior given the first bit. Under decoded feedback, which transitions are observed depends on the noise through the decoding outcome; the update ignores this observation mechanism and therefore defines an approximate posterior. [Section 7](#) analyzes the resulting estimation bias.

The initial knowledge is an estimate $\hat{\pi}_1$ of the error rate, with nothing known about memory. The prior sets $\alpha_i = \kappa \hat{\pi}_1$ and $\beta_i = \kappa(1 - \hat{\pi}_1)$ for $i = 0, 1$, so both parameters start at $\hat{\pi}_1$: a memoryless channel. The prior strength κ counts pseudo-transitions. After N_i observed transitions out of state i , of which N_{i1} lead to state 1, the posterior mean is

$$\hat{\theta}_i = \frac{\kappa}{\kappa + N_i} \hat{\pi}_1 + \frac{N_i}{\kappa + N_i} \frac{N_{i1}}{N_i}, \quad (8)$$

a weighted average of the prior mean and the empirical frequency, and the posterior standard deviation of θ_i is about $\sqrt{\theta_i(1 - \theta_i)/(\kappa + N_i)}$. With The prior on θ_i thus carries the weight of about $\kappa/(\pi_i M)$ frames of observations, which is small for p and large for q .

4.3 Decoder

Before each frame the receiver sorts the $O(n^2)$ pattern classes of [Lemma 2](#) by their log-probability (3) under $\hat{\theta}$ for strategy A, or under $\hat{\theta}^{(D)}$ for B, and queries the tie groups in that order. At the memoryless prior this is Hamming-weight GRAND. As $\hat{q}$ grows, the cost γ_b of a further error within a burst falls, and long single bursts move ahead of scattered errors ([Fig. 2](#)).

Algorithm 2 BBE on the two-state channel, frame t

- 1: $\hat{\theta} \leftarrow (\alpha_0/(\alpha_0 + \beta_0), \alpha_1/(\alpha_1 + \beta_1))$
 - 2: Receiver sorts the classes by $\log \mathbb{P}_{\hat{\theta}}$ for A and by $\log \mathbb{P}_{\hat{\theta}^{(D)}}$ for B
 - 3: Draw $\tilde{p} \sim \text{Beta}(\alpha_0, \beta_0)$, $\tilde{q} \sim \text{Beta}(\alpha_1, \beta_1)$; set $b_a \leftarrow \text{BLER}_a(\tilde{\theta}; \hat{\theta})$ by (5)
 - 4: **if** BBE-noise **then**
 - 5: $a_t \leftarrow \arg \min_a b_a + c \mathbf{1}\{a = B\}$
 - 6: **else** $\triangleright$ BBE-ACK: $m_0 = 0$; BBE-hybrid: $m_0 > 0$
 - 7: Draw $\tilde{e}_a \sim \text{Beta}(1 + L_a + m_0 b_a, 1 + S_a + m_0(1 - b_a))$ for each a
 - 8: $a_t \leftarrow \arg \min_a \tilde{e}_a + c \mathbf{1}\{a = B\}$
 - 9: **end if**
 - 10: Announce a_t in the header, send the frame with strategy a_t ; GRAND decodes each codeword
 - 11: Count transitions (6) over adjacent known bits and update the posterior by (7)
 - 12: Update the acknowledgment counts L_a, S_a by (9)
-

4.4 Selecting the encoding

The cost (2) combines the BLER with the delay penalty. The three variants of Section 2 estimate the BLER differently.

BBE-noise draws $\tilde{p} \sim \text{Beta}(\alpha_0, \beta_0)$ and $\tilde{q} \sim \text{Beta}(\alpha_1, \beta_1)$, and evaluates $J_a(\tilde{\theta}; \hat{\theta})$ with (5). In this evaluation the order of the tie groups, and hence A_g and L_g , comes from the decoder belief $\hat{\theta}$, and the group probabilities $\mathbb{P}_{\hat{\theta}}(Z \in g)$ come from the sample $\tilde{\theta}$, both mapped by Lemma 1 for B. It plays the strategy with the lower cost. Early on the samples spread widely, and as the counts grow they concentrate at $\hat{\theta}$. The greedy version uses $\tilde{\theta} = \hat{\theta}$.

BBE-ACK keeps, for each strategy a , discounted counts L_a and S_a of lost and decoded codewords. After a frame with strategy a_t and ℓ_t of D codewords lost,

$$L_a \leftarrow \gamma L_a + \ell_t \mathbf{1}\{a = a_t\}, \quad S_a \leftarrow \gamma S_a + (D - \ell_t) \mathbf{1}\{a = a_t\} \quad (9)$$

for both a , with discount $\gamma < 1$, because the loss rate of each strategy falls as the decoder learns. It draws $\tilde{e}_a \sim \text{Beta}(1 + L_a, 1 + S_a)$ and plays $\arg \min_a \tilde{e}_a + c \mathbf{1}\{a = B\}$.

BBE-hybrid draws $\tilde{\theta}$ as BBE-noise does, predicts $b_a = \text{BLER}_a(\tilde{\theta}; \hat{\theta})$, and adds m_0 pseudo-codewords to the counts of BBE-ACK, $\tilde{e}_a \sim \text{Beta}(1 + L_a + m_0 b_a, 1 + S_a + m_0(1 - b_a))$. When a strategy has few recent acknowledgments, the model's prediction dominates its sample, and when it has many, the acknowledgments do.

Any variant can be preceded by an *initial interleaving phase* of T_0 frames, in which every frame is interleaved. In the simulations we compare with fixed designs that always send plain or always interleave, with the learning decoder, and with a conventional link that interleaves every frame and keeps the memoryless decoder.

5 Interleaving and the Decoder's Belief

When the decoder uses belief $\hat{\theta}$, interleaving has the lower cost iff c is below the *critical penalty*

$$c^*(\theta, \hat{\theta}) = \text{BLER}_A(\theta; \hat{\theta}) - \text{BLER}_B(\theta; \hat{\theta}). \quad (10)$$

We write c_{mem}^* for the memoryless initial belief and c_{mat}^* for the matched belief $\hat{\theta} = \theta$. The first result concerns matched decoding as the block length grows.

Proposition 2.

- (i) $H(\theta^{(D)}) \geq H(\theta)$, with equality iff $\lambda = 0$ (for $D \geq 2$).
- (ii) Let $n \rightarrow \infty$ with $Q = 2^{nR_Q}$ and $r/n \rightarrow R_r > R_Q$. With a matched decoder, $\text{BLER}_A \rightarrow 0$ if $R_Q > H(\theta)$. Under any query order, $\text{BLER}_B \rightarrow 1$ if $R_Q < H(\theta^{(D)})$, and $\text{BLER}_A \rightarrow 1$ if $R_Q < H(\theta)$.
- (iii) For every $c > 0$ and every $R_Q < R_r$ with $R_Q \notin \{H(\theta), H(\theta^{(D)})\}$, strategy A with a matched decoder is optimal for all large n .

Proof. (i) By the Markov property, and because conditioning cannot increase entropy, $H(Z_t | Z_{t-D}) \geq H(Z_t | Z_{t-1}, Z_{t-D}) = H(Z_t | Z_{t-1})$. The left side is $H(\theta^{(D)})$ by Lemma 1. Equality holds iff $Z_t \perp Z_{t-1} | Z_{t-D}$. Since $0 < p, q < 1$, all entries of P^{D-1} are positive, so given Z_{t-D} both values of Z_{t-1} have positive probability. Conditional independence then requires $\mathbb{P}(Z_t = 1 | Z_{t-1} = b)$ not to depend on b , that is, $p = q$.

(ii) For a matched decoder, every pattern queried before Z is at least as likely as Z , so $\text{rank}(Z) \mathbb{P}_\theta(Z) \leq 1$. Hence $\mathbb{P}(\text{rank}(Z) > Q) \leq \mathbb{P}(-\frac{1}{n} \log_2 \mathbb{P}_\theta(Z) > R_Q)$, which tends to zero for $R_Q > H(\theta)$ by the Shannon–McMillan–Breiman theorem for ergodic Markov chains [15]. The probability of a false hit among the first Q queries is at most $Q2^{-r} \rightarrow 0$ because $R_Q < R_r$. For the converse, let $\mathcal{W}$ be the set of the first Q patterns of any fixed order, and let H' be the entropy rate of the codeword noise. For $\epsilon > 0$, $\mathbb{P}(Z \in \mathcal{W}) \leq \mathbb{P}(\mathbb{P}(Z) > 2^{-n(H'-\epsilon)}) + Q2^{-n(H'-\epsilon)}$, and both terms vanish when $R_Q < H' - \epsilon$. Correct decoding requires $Z \in \mathcal{W}$. Apply this with $H' = H(\theta^{(D)})$ for B (by Lemma 1) and $H' = H(\theta)$ for A.

(iii) If $H(\theta) < R_Q < H(\theta^{(D)})$, (ii) gives $J_A \rightarrow 0$ and $J_B \rightarrow 1 + c$. If $R_Q > H(\theta^{(D)})$, both BLERs vanish (apply (ii) to B with its own matched decoder), and $J_A \rightarrow 0 < c \leq \lim J_B$. If $R_Q < H(\theta)$, both BLERs tend to one, and $J_A \rightarrow 1 < 1 + c$. $\square$

Part (i) says that interleaving makes the noise of each codeword less predictable. A matched decoder needs about 2^{nH} queries, so for long codes interleaving increases the number of queries needed, and with the delay penalty it is the worse choice (part (iii)). At $n = 64$ a memoryless decoder can still benefit from interleaving, because interleaving reduces the variance of the number of errors in a codeword.

Proposition 3. Consider a memoryless belief with error rate below $1/2$ and ignore false hits. Whether a codeword is decoded depends on its noise only through the block weight $W = \sum_{i=1}^n Z_i$: weights below some w_Q always decode, weight w_Q decodes with some probability, and larger weights never do. For both strategies $\mathbb{E}W = n\pi_1$, and

$$\text{Var } W = \pi_0 \pi_1 \left[n + 2 \sum_{k=1}^{n-1} (n-k) \varphi^k \right], \quad \varphi = \lambda \text{ (A)}, \quad \varphi = \lambda^D \text{ (B)}. \quad (11)$$

For $\lambda > 0$ the variance is strictly smaller under B.

Proof. Under the memoryless belief the tie groups are the weights, queried in increasing order. By Lemma 1, $\mathbb{P}(Z_{i+k} = 1 | Z_i = 1) = \pi_1 + \pi_0 \varphi^k$, so $\text{Cov}(Z_i, Z_{i+k}) = \pi_0 \pi_1 \varphi^k$. Summing the covariances gives the formula, which is increasing in $\varphi \geq 0$, and $\lambda^D < \lambda$ for $0 < \lambda < 1$ and $D \geq 2$. $\square$

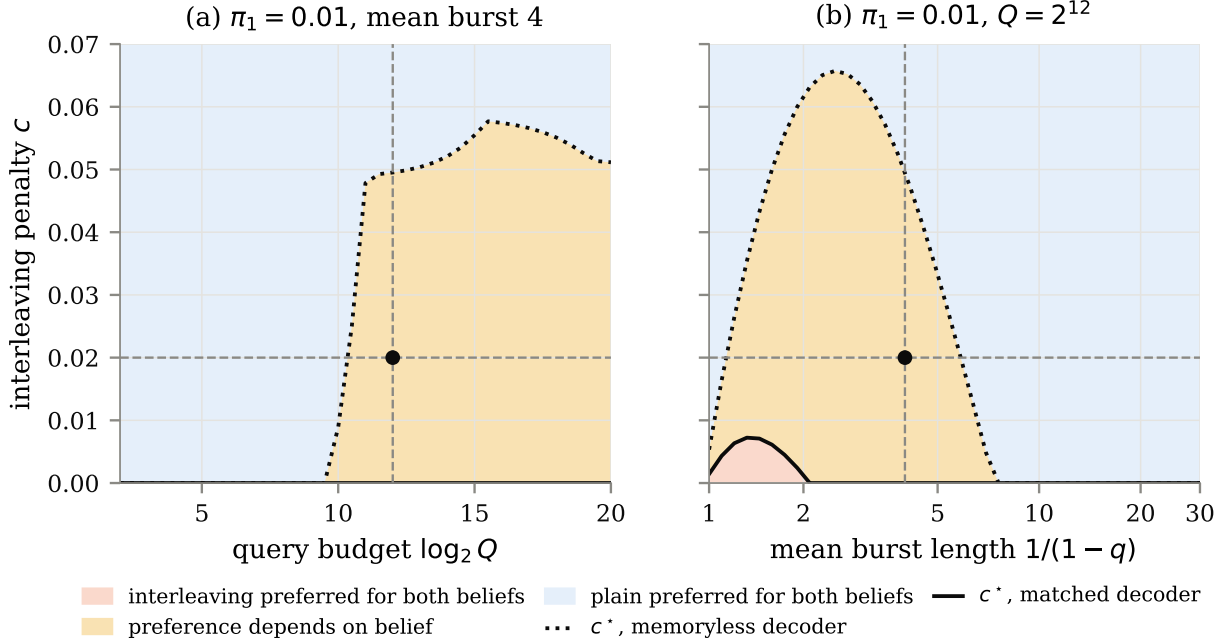


Figure 4: Critical penalties for the memoryless decoder (dotted) and the matched decoder (solid), clipped at zero, for $\pi_1 = 0.01$, RLC(64, 44) and $D = 8$, versus the query budget (left, mean burst length 4) and the mean burst length (right, $Q = 2^{12}$). The dot marks the operating point of Section 9.

When the budget covers more than the mean weight, the smaller variance under B leaves fewer blocks beyond the budget. In a normal approximation the memoryless decoder fails with probability about $\bar{\Phi}((w_Q + \frac{1}{2} - n\pi_1)/\sqrt{\text{Var } W})$, where $\bar{\Phi}$ is the standard normal tail function.

Proposition 2 is asymptotic. At short lengths the block entropy nH can lie far below $\log_2 Q$, and the BLER is then set by rare heavy blocks. Figure 4 shows both critical penalties at this length, computed from (5). Where $c < c_{\text{mat}}^*$, interleaving is preferred even with a matched decoder, which happens for short bursts. Where $c > c_{\text{mem}}^*$, plain transmission is preferred even with the memoryless decoder. Between the two curves the preferred strategy changes as the decoder learns the channel. The band is wide only when the budget covers long single bursts and the bursts are short enough for a depth- D interleaver to spread them.

6 Learning under Full Feedback

This section assumes that the learner observes the full noise sequence of every frame and counts all adjacent transitions in it. The observations then do not depend on the strategy played. We first show that the decoder reaches the oracle query order, then bound the regret of the selector, and finally approximate the frame at which the plain strategy becomes cheaper.

6.1 The decoder

For a belief θ' and strategy a , let $\mathcal{G}_a(\theta')$ be the sequence of tie groups, each viewed as a set of classes, that GRAND queries within the budget, up to and including the group that contains query Q . By

Proposition 1. $\text{BLER}_a(\theta''; \theta')$ depends on θ' only through $\mathcal{G}_a(\theta')$.

Definition 1. Write $\ell_\omega(\theta')$ for the log-probability (3) of a pattern in class ω under θ' . A parameter θ is *regular* if, for each strategy, any two classes that tie at the codeword-level parameter (θ for A, $\theta^{(D)}$ for B) have identical log-probability functions, $\ell_\omega \equiv \ell_{\omega'}$.

Regularity rules out ties that hold only at the particular parameter and keeps those that hold for every parameter. The non-regular parameters lie in the zero sets of finitely many analytic functions that are not identically zero (for B, composed with $\theta \mapsto \theta^{(D)}$, whose image has nonempty interior), so they form a set of Lebesgue measure zero.

Lemma 4. *If θ is regular, there are $\bar{\rho}_0, \bar{\rho}_1 > 0$ such that $\mathcal{G}_a(\theta') = \mathcal{G}_a(\theta)$ for both strategies whenever $|p' - p| < \bar{\rho}_0$ and $|q' - q| < \bar{\rho}_1$. In particular $\text{BLER}_a(\theta''; \theta') = \text{BLER}_a(\theta''; \theta)$ for every θ'' .*

Proof. Each ℓ_ω is continuous on $(0, 1)^2$, and so is the map $\theta' \mapsto \theta'^{(D)}$. There are finitely many classes. At a regular θ , any two classes either have identical functions, and then tie for all θ' , or differ at θ , and then keep the same strict order in a neighborhood. On the intersection of these neighborhoods the partition into tie groups and their order are unchanged, and so is $\mathcal{G}_a$. $\square$

We call the set of beliefs θ' with $\mathcal{G}_a(\theta') = \mathcal{G}_a(\theta)$ for both strategies the *ordering cell* of θ . A consistent learner makes the decoder coincide with the matched decoder after finitely many frames. The size of the cell depends on the channel and the budget, and it can be narrow, because many classes must keep their order within the budget. Proposition 4 bounds the loss of a belief outside the cell.

Proposition 4. *Let $\mathcal{S}$ be the set of patterns that either the matched decoder or the decoder with belief $\hat{\theta}$ can query within the budget, and let $\eta = \max_{z \in \mathcal{S}} |\log \mathbb{P}_{\hat{\theta}}(z) - \log \mathbb{P}_{\theta}(z)|$, where for strategy B both probabilities are taken under the codeword-level parameters $\hat{\theta}^{(D)}$ and $\theta^{(D)}$. Then*

$$0 \leq \text{BLER}(\theta; \hat{\theta}) - \text{BLER}(\theta; \theta) \leq (1 - e^{-2\eta})(1 - \text{BLER}(\theta; \theta)) \leq 2\eta. \quad (12)$$

Proof. Fix any realization of the random tie-breaking of each decoder, and let z_k and z_k^* be the k -th patterns queried by the decoder with belief $\hat{\theta}$ and by the matched decoder. The success probability is $\sum_k w_k \mathbb{P}_{\theta}(z_k)$ with $w_k = (1 - \varepsilon)^{k-1} \mathbf{1}\{k \leq Q\}$, a nonincreasing sequence. By the rearrangement inequality the matched order maximizes it, which gives the left inequality. For the right one it suffices to consider $k \leq Q$, since $w_k = 0$ otherwise. Then $\mathbb{P}_{\hat{\theta}}(z_k)$ is the k -th largest value of $\mathbb{P}_{\hat{\theta}}$. Each of the k patterns $z_1^*, \dots, z_k^*$ lies in $\mathcal{S}$ and satisfies $\mathbb{P}_{\hat{\theta}}(z_j^*) \geq e^{-\eta} \mathbb{P}_{\theta}(z_j^*) \geq e^{-\eta} \mathbb{P}_{\theta}(z_k^*)$, so $\mathbb{P}_{\hat{\theta}}(z_k) \geq e^{-\eta} \mathbb{P}_{\theta}(z_k^*)$. Since $z_k \in \mathcal{S}$ as well, $\mathbb{P}_{\theta}(z_k) \geq e^{-\eta} \mathbb{P}_{\hat{\theta}}(z_k) \geq e^{-2\eta} \mathbb{P}_{\theta}(z_k^*)$. Multiplying by w_k and summing gives $1 - \text{BLER}(\theta; \hat{\theta}) \geq e^{-2\eta}(1 - \text{BLER}(\theta; \theta))$. The matched success probability does not depend on its tie-breaking, and averaging over the tie-breaking of the other decoder completes the proof. $\square$

The patterns in $\mathcal{S}$ have few bursts, and η is small when $\hat{\theta}$ is close to θ in relative terms, so beliefs well outside the ordering cell can still decode almost as well as the matched decoder.

6.2 Bounded regret

Let $\mathcal{F}_{t-1}$ be the information after $t - 1$ frames. The decoder belief $\hat{\theta}_t$ is the posterior mean given $\mathcal{F}_{t-1}$, and $\tilde{\theta}_t$ is drawn from the posterior given $\mathcal{F}_{t-1}$. The regret after T frames is

$$R_T = \sum_{t=1}^T \mathbb{E}[J_{a_t}(\theta; \hat{\theta}_t) - J^*(\theta)], \quad (13)$$

which includes both the cost of playing a suboptimal strategy and the cost of decoding with a mismatched belief. Write a^* for the optimal strategy and $B(\rho) = \{\theta' : |p' - p| < \rho_0, |q' - q| < \rho_1\}$ for a box of radii $\rho = (\rho_0, \rho_1)$.

Assumption 1. For some $\delta \geq 0$ there are radii $\rho_0, \rho_1 \in (0, 1]$ such that, for all $\hat{\theta}, \tilde{\theta} \in B(\rho)$,

- (a) $J_{a^*}(\theta; \hat{\theta}) \leq J^*(\theta) + \delta$, and
- (b) $\arg \min_a J_a(\tilde{\theta}; \hat{\theta}) = \{a^*\}$.

With $\delta = 0$, condition (a) holds inside the ordering cell of [Lemma 4](#). A positive δ permits a larger box, in line with [Proposition 4](#). The next lemma gives sufficient conditions for $\delta = 0$.

Lemma 5. If θ is regular and $c \neq c_{\text{mat}}^*$, [Assumption 1](#) holds with $\delta = 0$. Let $\Delta(\theta') = J_B(\theta'; \theta) - J_A(\theta'; \theta)$. Choose radii $\bar{\rho}_i \leq 1$ as in [Lemma 4](#), small enough that the closed box $\overline{B(\bar{\rho})}$ lies in $(0, 1)^2$, and let Λ_0, Λ_1 bound $|\partial\Delta/\partial p'|$ and $|\partial\Delta/\partial q'|$ on it. Then one can take $\rho_i = \min\{\bar{\rho}_i, |c - c_{\text{mat}}^*|/(2\Lambda_i)\}$, with $\rho_i = \bar{\rho}_i$ if $\Lambda_i = 0$.

Proof. For $\hat{\theta} \in B(\bar{\rho})$, [Lemma 4](#) gives $J_a(\theta''; \hat{\theta}) = J_a(\theta''; \theta)$ for all θ'' . With $\theta'' = \theta$ this yields (a) with $\delta = 0$. With $\theta'' = \tilde{\theta}$ it reduces (b) to the requirement that $\Delta(\tilde{\theta})$ have the sign of $\Delta(\theta) = c - c_{\text{mat}}^* \neq 0$. By (5), Δ is a finite sum of smooth functions of $\tilde{\theta}$ on the compact box, and $\theta' \mapsto \theta'^{(D)}$ is smooth, so $\Lambda_i < \infty$. The box is convex, and by the mean value theorem $|\Delta(\tilde{\theta}) - \Delta(\theta)| < \Lambda_0\rho_0 + \Lambda_1\rho_1 \leq |c - c_{\text{mat}}^*|$ for $\tilde{\theta} \in B(\rho)$, which fixes the sign. $\square$

Let $d_0 = |\hat{\pi}_1 - p|$ and $d_1 = |\hat{\pi}_1 - q|$ be the distances between the prior mean and the true parameters. By [Lemma 9](#) in [Section 8](#) there are constants $C_i \geq 1$ and $I_i > 0$, depending only on θ , such that the number $V_i(\nu)$ of visits to state i among the first ν bits satisfies $\mathbb{P}(V_i(\nu) < \nu\pi_i/2) \leq C_i e^{-I_i\nu}$.

The argument behind the theorem is simple. Once the posterior mean and the posterior sample both lie in the box of [Assumption 1](#), the decoder is within δ of the matched decoder and the sample ranks the two strategies correctly. Under full feedback each frame adds about $\pi_i M$ transitions out of state i whatever strategy is played, so the posterior concentrates at a rate that does not depend on the selector's choices. The expected number of frames spent outside the box is therefore finite, and it does not grow with the horizon. The theorem makes each of these steps quantitative.

Theorem 1. Assume full feedback, [Assumption 1](#) and $\kappa \geq 1$. For every horizon T , BBE-noise satisfies

$$\mathbb{E} \#\{t \leq T : J_{a_t}(\theta; \hat{\theta}_t) > J^*(\theta) + \delta\} \leq K, \quad (14)$$

where

$$K = K_0 + \sum_{i \in \{0,1\}} \frac{1}{\pi_i M} \left(\frac{6\kappa d_i}{\rho_i} + \frac{45}{\rho_i^2} \right), \quad K_0 = 12 + \sum_{i \in \{0,1\}} \left[\frac{2\kappa + 2}{\pi_i M} + C_i \left(1 + \frac{1}{I_i M} \right) \right]. \quad (15)$$

Consequently $R_T \leq \delta T + (1 + c)K$, and the regret is bounded uniformly in T when the margin holds with $\delta = 0$. The same bounds hold for the greedy selector $\tilde{\theta}_t = \hat{\theta}_t$.

[Section 8](#) contains the proof. Combined with [Lemma 5](#), the theorem shows how the bound depends on the problem parameters.

Corollary 1. *If θ is regular, $c \neq c_{\text{mat}}^*$ and $\kappa \geq 1$, then under full feedback*

$$\sup_T R_T \leq (1+c) \left[K_0 + \sum_i \frac{1}{\pi_i M} \left(\frac{6\kappa d_i}{\rho_i} + \frac{45}{\rho_i^2} \right) \right], \quad \rho_i = \min \left\{ \bar{\rho}_i, \frac{|c - c_{\text{mat}}^*|}{2\Lambda_i} \right\}, \quad (16)$$

where K_0 does not depend on c or on the margin.

The bound grows at most as $|c - c_{\text{mat}}^*|^{-2}$ when the penalty approaches the critical penalty of the matched decoder. The margin-dependent terms are inversely proportional to the number $\pi_i M$ of transitions out of state i observed per frame, which favors the error-state term, but the radii matter as well. They grow linearly in $\kappa d_i / \rho_i$, the prior strength times the prior's distance from the truth relative to the margin. The budget Q enters through c_{mat}^* , through Λ_i , and through the cell size $\bar{\rho}_i$, which shrinks as more classes have to be ordered correctly.

In a standard stochastic bandit an arm's reward is observed only when the arm is played, and every consistent policy has regret of order $\log T$ [20]. Under full feedback all strategies are functions of one parameter that the channel reveals in every frame. The greedy selector, which never explores, therefore has bounded regret as well, as in structured bandits with a shared parameter [13, 14]. The argument uses three properties, namely that the observations do not depend on the action, that the posterior concentrates at an exponential rate, and that a margin exists.

The constants are conservative. With $\delta = 0$ the radius ρ_0 cannot exceed the width of the ordering cell in p , which enters the bound as $1/\rho_0^2$, and Hoeffding's inequality does not use the fact that p is small. A positive δ allows much larger boxes by Proposition 4. Section 9.2 evaluates the constants at the operating point of the simulations.

6.3 When the plain strategy becomes cheaper

Replacing the counts in the posterior by their expectations gives a deterministic approximation of the posterior mean. With an exact prior ($\hat{\pi}_1 = \pi_1$), after t frames of full feedback,

$$\bar{p}_t = \frac{\kappa\pi_1 + p\pi_0 tM}{\kappa + \pi_0 tM}, \quad \bar{q}_t = \frac{\kappa\pi_1 + q\pi_1 tM}{\kappa + \pi_1 tM}. \quad (17)$$

Proposition 5. *Let $\tau = \min\{t : J_A(\theta; \bar{\theta}_t) \leq J_B(\theta; \bar{\theta}_t)\}$. Then (i) $\tau = 0$ iff $c \geq c_{\text{mem}}^*$; (ii) if θ is regular and $c > c_{\text{mat}}^*$, then $\tau < \infty$, and if $c < c_{\text{mat}}^*$, the plain strategy is eventually more expensive; (iii) τ is nonincreasing in c ; (iv) $\tau = \min\{t : tM/\kappa \in X\}$ for a set $X \subset [0, \infty)$ that depends on (θ, c, Q) but not on κ or M , and if X is a ray $[x^*, \infty)$, then $\tau = \lceil x^* \kappa / M \rceil$.*

Proof. The approximation depends on (t, κ, M) only through $x = tM/\kappa$. Taking $X = \{x : J_A(\theta; \bar{\theta}(x)) \leq J_B(\theta; \bar{\theta}(x))\}$ gives (iv). The difference $J_A - J_B$ equals $c^*(\theta, \bar{\theta}_t) - c$. It is decreasing in c , which gives (iii), and equals $c_{\text{mem}}^* - c$ at $t = 0$, which gives (i). As $t \rightarrow \infty$, $\bar{\theta}_t \rightarrow \theta$, and by Lemma 4 the difference equals $c_{\text{mat}}^* - c$ for all large t , which gives (ii). $\square$

The approximation after t frames corresponds to $\hat{\theta}_{t+1}$ in Theorem 1.

7 Learning under Decoded Feedback

With decoded feedback the receiver learns only from codewords it decodes. Decoding fails most often on the patterns that the current decoder ranks low, so the observations lean toward what the

decoder already expects, and their content depends on the strategy played. This section bounds the resulting bias and characterizes the long-run estimate. The strategy determines how fast the estimate gets there.

Consider a plain codeword, let N_{ij} be the number of $i \rightarrow j$ transitions within it, and let $N_{1\cdot} = N_{10} + N_{11}$, let $\mathcal{D}$ be the event that it is decoded, and let $\varepsilon_{\text{dec}} = \mathbb{P}(\mathcal{D}^c)$ be its BLER. Counts accumulated from decoded codewords estimate $q_c = \mathbb{E}[N_{11}\mathbf{1}_{\mathcal{D}}]/\mathbb{E}[N_{1\cdot}\mathbf{1}_{\mathcal{D}}]$.

Lemma 6.

$$q_c - q = -\frac{\mathbb{E}[(N_{11} - qN_{1\cdot})\mathbf{1}_{\mathcal{D}^c}]}{\mathbb{E}[N_{1\cdot}\mathbf{1}_{\mathcal{D}}]}, \quad |q_c - q| \leq \frac{\sqrt{q(1-q)(n-1)\pi_1\varepsilon_{\text{dec}}}}{\mathbb{E}[N_{1\cdot}\mathbf{1}_{\mathcal{D}}]}, \quad (18)$$

and analogously for p with π_0 and $N_{0\cdot} = N_{00} + N_{01}$.

Proof. The process $X_k = \sum_{s \leq k} \mathbf{1}\{Z_s = 1\}(Z_{s+1} - q)$ is a martingale with respect to the natural filtration, so $\mathbb{E}[N_{11} - qN_{1\cdot}] = \mathbb{E}X_n = 0$, which gives the identity after splitting on $\mathcal{D}$. Its increments are orthogonal with $\mathbb{E}[\mathbf{1}\{Z_s = 1\}(Z_{s+1} - q)^2] = \pi_1 q(1-q)$, so $\mathbb{E}X_n^2 = q(1-q)(n-1)\pi_1$. The Cauchy–Schwarz inequality gives $|\mathbb{E}[X_n\mathbf{1}_{\mathcal{D}^c}]| \leq \sqrt{\mathbb{E}X_n^2} \sqrt{\varepsilon_{\text{dec}}}$. $\square$

The bound decreases with the BLER of the decoder that produced the data. The long-run estimate of a learner that always sends plain frames solves $\hat{\theta} = (p_c, q_c)(\hat{\theta})$, where the right side can be computed exactly by weighting each class with its probability and its decoding probability. Iterating this map from the memoryless prior gives the long-run estimate. The fixed point need not lie in the ordering cell, but by [Proposition 4](#) its decoder loses little when it is close to θ in relative terms.

The fixed point describes the limit, and the strategy determines the speed of approach. Interleaved codewords see little memory and are often decoded when the channel produced a long burst, and after de-interleaving they return the burst statistics that lost plain codewords would have carried. Extending [Theorem 1](#) to decoded feedback requires accounting for this dependence of the observations on the strategy. [Section 9](#) measures its effect.

8 Proof of Theorem 1

The proof has five steps. Step 1 shows that the transitions out of each state are i.i.d. Bernoulli variables, and Step 2 bounds their running averages uniformly. Step 3 shows that each state is visited often enough. Step 4 relates the posterior mean and the posterior samples to the empirical frequencies. Step 5 combines these bounds to show that frames in which both the belief and the sample lie in the margin box are δ -optimal, and sums the probabilities of the remaining frames.

Notation. Before frame t the learner has observed $\nu_t = (t-1)M$ consecutive bits $Z_1, \dots, Z_{\nu_t}$. For $i \in \{0, 1\}$ let $N_i = \#\{s < \nu_t : Z_s = i\}$ be the number of observed transitions out of state i , and let N_{i1} be the number of those followed by a one. Write $\theta_0 = p$, $\theta_1 = q$, and $\check{\theta}_i = N_{i1}/N_i$ for the empirical transition frequency (defined when $N_i \geq 1$). Write $\hat{\theta}_{t,i}$ and $\tilde{\theta}_{t,i}$ for the i -th components of $\hat{\theta}_t$ and $\tilde{\theta}_t$.

Step 1: successors are i.i.d.

Lemma 7. Fix i , let $\sigma_1 < \sigma_2 < \dots$ be the successive times $s \geq 1$ with $Z_s = i$, and let $\xi_k = Z_{\sigma_k+1}$. Then $\xi_1, \xi_2, \dots$ are i.i.d. Bernoulli(θ_i), and $N_{i1} = \sum_{k \leq N_i} \xi_k$.

Proof. The chain is irreducible and recurrent, so every σ_k is finite almost surely, and each σ_k is a stopping time for the filtration $\mathcal{Z}_s = \sigma(Z_1, \dots, Z_s)$. By the strong Markov property, conditional on $\mathcal{Z}_{\sigma_k}$ the next bit Z_{σ_k+1} is Bernoulli(θ_i). The variables $\xi_1, \dots, \xi_{k-1}$ are $\mathcal{Z}_{\sigma_k}$ -measurable because $\sigma_j + 1 \leq \sigma_k$ for $j < k$. Hence $\mathbb{P}(\xi_1 = b_1, \dots, \xi_k = b_k) = \mathbb{P}(\xi_1 = b_1, \dots, \xi_{k-1} = b_{k-1}) \theta_i^{b_k} (1 - \theta_i)^{1-b_k}$, and induction on k gives independence. The identity for N_{i1} holds by definition. $\square$

Step 2: a maximal inequality for running averages

Because N_i is random, we need a deviation bound that holds simultaneously for all sample sizes beyond a threshold.

Lemma 8. *Let $\xi_1, \xi_2, \dots$ be i.i.d. with values in $[0, 1]$ and mean μ , and let $\bar{\xi}_j = \frac{1}{j} \sum_{k \leq j} \xi_k$. For every $m \geq 1$ and $x > 0$,*

$$\mathbb{P}\left(\sup_{j \geq m} |\bar{\xi}_j - \mu| \geq x\right) \leq 2e^{-2mx^2}. \quad (19)$$

Proof. Let $\mathcal{H}_j = \sigma(\bar{\xi}_j, \bar{\xi}_{j+1}, \bar{\xi}_{j+2}, \dots)$, a decreasing family. By exchangeability, $\mathbb{E}[\xi_k | \mathcal{H}_{j+1}] = \bar{\xi}_{j+1}$ for $k \leq j+1$, so $\mathbb{E}[\bar{\xi}_j | \mathcal{H}_{j+1}] = \bar{\xi}_{j+1}$: the averages form a reverse martingale. For $\zeta > 0$, $Y_j = e^{\zeta(\bar{\xi}_j - \mu)}$ is a reverse submartingale by Jensen's inequality. For $L \geq m$, Doob's maximal inequality applied to $(Y_L, Y_{L-1}, \dots, Y_m)$, a submartingale in the order written, gives $\mathbb{P}(\max_{m \leq j \leq L} \bar{\xi}_j - \mu \geq x) \leq e^{-\zeta x} \mathbb{E} Y_m$. By Hoeffding's lemma, $\mathbb{E} Y_m \leq e^{\zeta^2/(8m)}$. Choosing $\zeta = 4mx$ gives e^{-2mx^2} , and letting $L \rightarrow \infty$ gives the one-sided bound. The lower tail is symmetric. $\square$

Step 3: enough visits

Lemma 9. *Let $V_i(v) = \#\{s \leq v : Z_s = i\}$. For each $i \in \{0, 1\}$ there are constants $C_i \geq 1$ and $I_i > 0$, depending only on θ , such that for all $v \geq 1$, $\mathbb{P}(V_i(v) < v\pi_i/2) \leq C_i e^{-I_i v}$.*

Proof. Fix i . For $s \geq 0$ let Γ_s be the diagonal matrix with $(\Gamma_s)_{ii} = e^{-s}$ and $(\Gamma_s)_{jj} = 1$ for $j \neq i$. Summing over paths, $\mathbb{E} e^{-sV_i(v)} = \pi^\top \Gamma_s (P\Gamma_s)^{v-1} \mathbf{1}$. The matrix $A_s = P\Gamma_s$ has positive entries, so by the Perron–Frobenius theorem it has a simple largest eigenvalue $\psi(s) > 0$ with a positive right eigenvector v_s . Since $\mathbf{1} \leq v_s / \min v_s$ entrywise and A_s has nonnegative entries, $(A_s)^{v-1} \mathbf{1} \leq \psi(s)^{v-1} v_s / \min v_s \leq \psi(s)^{v-1} \chi_s \mathbf{1}$ with $\chi_s = \max v_s / \min v_s$. Hence, as $\pi^\top \Gamma_s \mathbf{1} \leq 1$, $\mathbb{E} e^{-sV_i(v)} \leq \chi_s \psi(s)^{v-1}$, and by Markov's inequality

$$\mathbb{P}(V_i(v) < v\pi_i/2) \leq e^{sv\pi_i/2} \mathbb{E} e^{-sV_i(v)} \leq \frac{\chi_s}{\psi(s)} \exp(-v\phi_i(s)), \quad \phi_i(s) = -\frac{s\pi_i}{2} - \log \psi(s). \quad (20)$$

Because $\psi(s)$ is a simple eigenvalue it is differentiable, with $\psi(0) = 1$ and, by first-order perturbation with left eigenvector $\pi^\top$ and right eigenvector $\mathbf{1}$ of $A_0 = P$,

$$\psi'(0) = \frac{\pi^\top P \Gamma'_0 \mathbf{1}}{\pi^\top \mathbf{1}} = -\pi^\top P e_i = -\pi_i. \quad (21)$$

Hence $\phi_i(s) = s\pi_i/2 + o(s)$, and there is $s_i > 0$ with $\phi_i(s_i) > 0$. Take $I_i = \phi_i(s_i)$ and $C_i = \max\{1, \chi_{s_i}/\psi(s_i)\}$. $\square$

Step 4: the posterior mean and the posterior samples

The learner's posterior of θ_i given $\mathcal{F}_{t-1}$ (conditioned on the first bit, as in [Lemma 3](#)) is $\text{Beta}(\alpha_i + N_{i1}, \beta_i + N_i - N_{i1})$ with $\alpha_i + \beta_i = \kappa$ and prior mean $\alpha_i/\kappa = \hat{\pi}_1$. Its mean $\mu_i = (\alpha_i + N_{i1})/(\kappa + N_i)$, which is the component $\hat{\theta}_{t,i}$ of the decoder belief, satisfies

$$|\mu_i - \check{\theta}_i| = \frac{\kappa |\hat{\pi}_1 - \check{\theta}_i|}{\kappa + N_i} \leq \frac{\kappa (d_i + |\check{\theta}_i - \theta_i|)}{N_i}. \quad (22)$$

A $\text{Beta}(\alpha, \beta)$ variable is sub-Gaussian with variance proxy $1/(4(\alpha + \beta + 1))$ [16]. Hence, conditionally on $\mathcal{F}_{t-1}$,

$$\mathbb{P}(|\check{\theta}_{t,i} - \mu_i| \geq x \mid \mathcal{F}_{t-1}) \leq 2e^{-2(\kappa + N_i + 1)x^2} \leq 2e^{-2N_ix^2}. \quad (23)$$

Step 5: assembling the bound

Fix $t \geq 2$ and set $x_i = \rho_i/3$ and $m_{i,t} = v_t \pi_i/2 - 1$. Since j ranges over integers, the supremum over $j \geq m_{i,t}$ equals the supremum over $j \geq \lceil m_{i,t} \rceil$, and [Lemma 8](#) gives the bound $2e^{-2m_{i,t}x_i^2}$. Consider the events

$$E_V = \{V_i(v_t) \geq v_t \pi_i/2 \text{ for } i = 0, 1\},$$

$$E_i = \left\{ \sup_{j \geq m_{i,t}} |\bar{\xi}_j^{(i)} - \theta_i| < x_i \right\}, \quad E'_i = \{|\check{\theta}_{t,i} - \mu_i| < x_i\},$$

where $\bar{\xi}_j^{(i)}$ are the running averages of the successors of state i ([Lemma 7](#)). On E_V we have $N_i \geq V_i(v_t) - 1 \geq m_{i,t}$, because at most the last observed bit has no observed successor. Suppose in addition that

$$m_{i,t} \geq \kappa(1 + 3d_i/\rho_i) \quad \text{for } i = 0, 1, \quad (24)$$

which, since $\kappa \geq 1$, also gives $m_{i,t} \geq 1$, so that $\check{\theta}_i$ is defined on E_V . Then on $E_V \cap E_i$ we have $|\check{\theta}_i - \theta_i| < \rho_i/3$, and by (22) $|\mu_i - \check{\theta}_i| \leq \kappa(d_i + \rho_i/3)/m_{i,t} \leq \rho_i/3$. Hence $|\hat{\theta}_{t,i} - \theta_i| < 2\rho_i/3$. On E'_i as well, $|\check{\theta}_{t,i} - \theta_i| < \rho_i$. When this holds for both i , both $\hat{\theta}_t$ and $\check{\theta}_t$ lie in $B(\rho)$. By [Assumption 1\(b\)](#) the selector plays a^* , and by (a) the frame costs at most $J^* + \delta$. Therefore

$$\mathbb{P}(J_{a_t}(\theta; \hat{\theta}_t) > J^* + \delta) \leq \mathbb{P}(E_V^c) + \sum_i [\mathbb{P}(E_V \cap E_i^c) + \mathbb{P}(E_V \cap E_i \cap E'_i{}^c)]. \quad (25)$$

By [Lemma 9](#), $\mathbb{P}(E_V^c) \leq \sum_i C_i e^{-I_i v_t}$. By [Lemma 8](#), $\mathbb{P}(E_i^c) \leq 2e^{-2m_{i,t}x_i^2}$. By (23) and $N_i \geq m_{i,t}$ on E_V , $\mathbb{P}(E_V \cap E'_i{}^c) \leq 2e^{-2m_{i,t}x_i^2}$. With $x_i \leq 1/3$,

$$4e^{-2m_{i,t}x_i^2} = 4e^{2x_i^2} e^{-v_t \pi_i x_i^2} \leq 5e^{-(t-1)u_i}, \quad u_i = \frac{\pi_i M \rho_i^2}{9}. \quad (26)$$

Condition (24) holds for all $t \geq t_0$, where $t_0 \leq 2 + \sum_i [2\kappa(1 + 3d_i/\rho_i) + 2]/(\pi_i M)$. Bounding the probability by one for $t < t_0$ and summing the geometric series $\sum_{t \geq 1} e^{-(t-1)u} \leq 1 + 1/u$ for $t \geq t_0$ gives

$$\mathbb{E} \#\{t \leq T : J_{a_t}(\theta; \hat{\theta}_t) > J^* + \delta\} \leq t_0 + \sum_i C_i \left(1 + \frac{1}{I_i M}\right) + \sum_i 5 \left(1 + \frac{9}{\pi_i M \rho_i^2}\right), \quad (27)$$

which, after inserting the bound on t_0 , is (15). A frame counted in (14) costs at most $1 + c$ more than J^* , and each remaining frame at most δ more, so $R_T \leq \delta T + (1 + c)K$. For the greedy selector the events E'_i are not needed, and the bound holds a fortiori. $\square$

9 Simulation Results

9.1 Setup

We use an RLC(64,44) ($r = 20$), $D = 8$ codewords per frame ($M = 512$ channel uses), budget $Q = 2^{12}$ and penalty $c = 0.02$. The channel has $\pi_1 = 0.01$ and mean burst length 4 ($p \approx 0.0025$, $q = 0.75$). Unless stated otherwise, the prior uses $\hat{\pi}_1 = \pi_1$ and $\kappa = 200$, which gives the prior on q the weight of about 40 frames of observations of the error state. Runs last 400 frames, results are averages over 20 runs, feedback is decoded, and control signaling is error free. BBE-ACK and BBE-hybrid use the discount $\gamma = 0.95$, and BBE-hybrid uses $m_0 = 20$ pseudo-codewords. The simulator counts transitions within frames ($M - 1$ per frame instead of the M assumed in Section 6), and the deterministic approximation uses the same convention. The plotted costs are expected costs computed from (5) for each decision, which removes decoding noise from the curves. The BLERs at this operating point are the $Q = 2^{12}$ columns of Table 2, and $J^* = 0.0097$. The penalty $c = 0.02$ lies between $c_{\text{mat}}^* = -0.020$ and $c_{\text{mem}}^* = 0.050$, in the band of Fig. 4 where the preferred strategy changes as the decoder learns.

9.2 The analysis at the operating point

At this operating point a frame contains about $\pi_0 M \approx 507$ transitions out of state 0 and $\pi_1 M \approx 5$ out of state 1, so with $\kappa = 200$ the prior on p is outweighed within the first frame, while the prior on q carries the weight of about 40 frames.

Interleaving. The block weight has mean 0.64 under both strategies and variance 4.15 under A and 0.77 under B (Proposition 3), which is why the memoryless decoder has BLER 0.089 under A and 0.039 under B. The block entropies $nH(\theta) \approx 2.1$ and $nH(\theta^{(8)}) \approx 5.0$ bits are far below $\log_2 Q = 12$.

Decoder. The ordering cell is $|\hat{p} - p| < 1.2 \times 10^{-5}$ by $|\hat{q} - q| < 1.7 \times 10^{-4}$. Beyond it the loss is small: evaluated on a grid, a belief within 10% of p and within 0.05 of q raises the BLER by at most 1.7×10^{-4} , 1.7% of J^* .

Regret bound. The visit constants of Lemma 9 are $(C_1, I_1) \approx (1.34, 2.5 \times 10^{-4})$ and $(C_0, I_0) \approx (11.4, 0.12)$, and the visit terms in (15) contribute about 23 frames. With $\delta = 0$ the radius ρ_0 is at most 1.2×10^{-5} , and the term $45/(\pi_0 M \rho_0^2)$ exceeds 6×10^8 frames, of which a factor of order $1/(4p) \approx 100$ comes from Hoeffding's inequality. The simulated regret under full feedback is 1.72 after 400 frames and grows by 0.10 over the next 1100 frames.

Crossing time. In Proposition 5, X is a ray with $x^* \approx 20$, so $\tau = 8$ frames for $\kappa = 200$ and $\tau = 40$ for $\kappa = 1000$.

Decoded feedback. For a learner that sends only plain frames, the Hamming decoder gives $q_c = 0.32$ in the first iteration of the map of Section 7, since it decodes few long bursts. The iteration converges to $(p_c, q_c) = (0.0022, 0.749)$, against the true $(0.0025, 0.75)$. This point lies outside the ordering cell, since its p differs by 3×10^{-4} , but its BLER equals the matched BLER to four digits.

9.3 BBE in operation

Figure 5 follows the loop. As the posterior learns the bursts, the decoder built from it lowers the cost of the plain strategy from 0.089 toward J^* (panel b). In the BBE-hybrid runs, the plain strategy becomes cheaper than interleaving after a median of 12 frames, against 8 frames predicted by

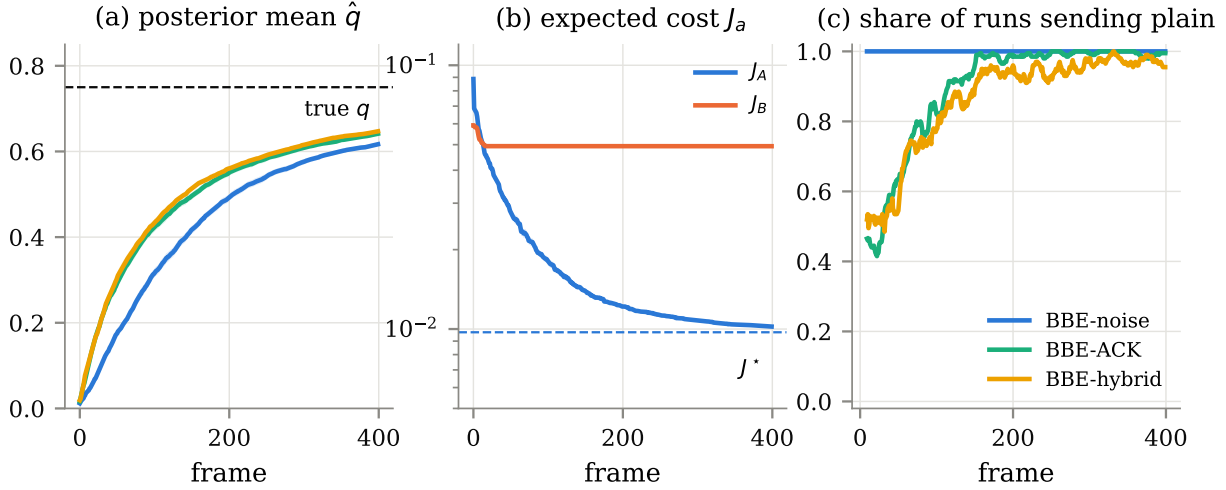


Figure 5: BBE at the operating point with decoded feedback (20 runs, mean $\pm$ one standard error). (a) Posterior mean of q . (b) Expected cost of each strategy for the decoder the receiver holds in that frame (BBE-hybrid runs); the dashed line is J^* . (c) Fraction of runs sending plain (10-frame moving average).

Proposition 5 for full feedback. By frame 400 the BLER of the plain strategy is within about 5% of J^* . The BLER of the interleaved strategy falls from 0.039 to 0.029 (its cost from 0.059 to 0.049) as the decoder learns the residual memory $\lambda^8 \approx 0.1$ of interleaved codewords.

BBE-ACK and BBE-hybrid send 48% and 47% of the first 50 frames interleaved (panel c), while plain frames are still losing codewords, and then move to plain transmission. The interleaved frames speed up learning: after 50 frames the posterior mean of q is 0.29 under BBE-ACK and 0.30 under BBE-hybrid, against 0.17 under BBE-noise (panel a).

BBE-noise sends plain from the first frame, although [Section 5](#) shows that interleaving is the better choice for the initial decoder. That comparison uses the true, bursty channel. BBE-noise compares the encodings on channels drawn from its posterior, which early on have almost no memory. Interleaving does not change the noise of a memoryless channel, so the delay penalty makes plain the cheaper choice. By the time the posterior has learned the memory, the decoder has learned it too, and plain is then the better choice on the true channel as well. BBE-noise therefore learns the bursts from decoded plain codewords alone. Where $c < c_{\text{mat}}^*$, as for short bursts in [Fig. 4](#), interleaving remains the better choice after learning, and BBE-noise interleaves once its posterior has learned the memory.

9.4 Regret

[Figure 6](#) and [Table 3](#) compare the variants for $\kappa = 200$ and for a more confident prior, $\kappa = 1000$. We split the regret of each frame into a *learning* part, $\min_a J_a(\theta; \hat{\theta}_t) - J^*$, which is the cost of a decoder that has not yet learned the channel, and a *selection* part, $J_{a_t}(\theta; \hat{\theta}_t) - \min_a J_a(\theta; \hat{\theta}_t)$, which is the cost of the worse strategy for the current decoder.

All BBE variants are far below the fixed interleaved designs. Over 400 frames the conventional link, which interleaves every frame and keeps the memoryless decoder, accumulates a regret of 19.8 and has an average cost of 0.059 per frame. BBE-ACK accumulates 3.91 with an average cost of 0.020, and BBE-hybrid 4.22. Interleaving every frame with the learning decoder accumulates 15.9,

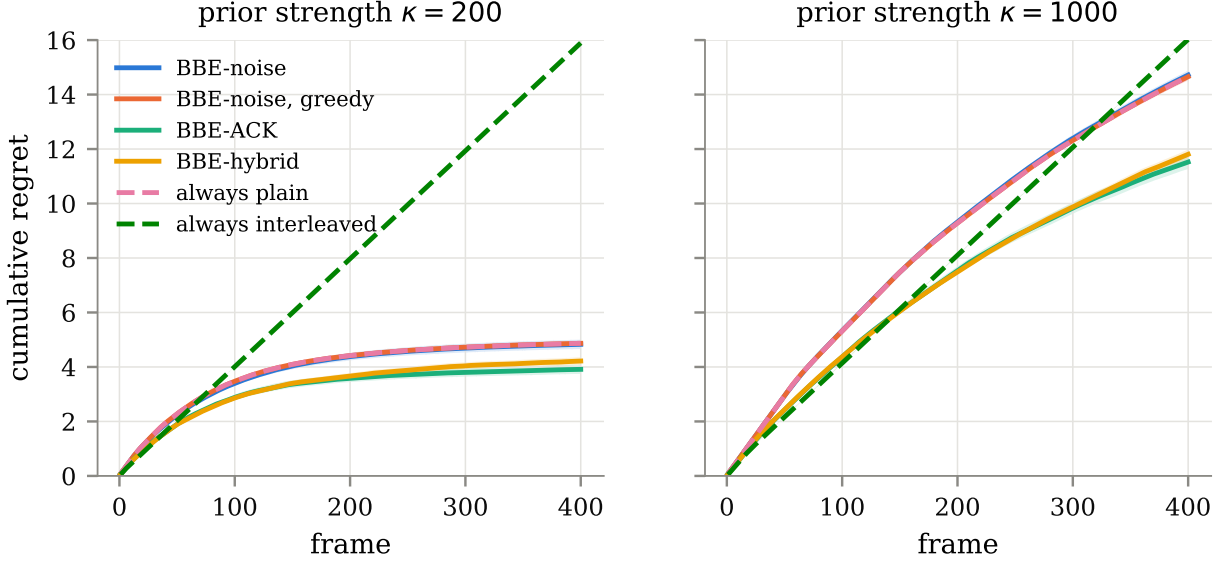


Figure 6: Cumulative regret with decoded feedback for the BBE variants and the fixed strategies (20 runs).

Table 3: Regret over 400 frames with decoded feedback, split into learning and selection parts (mean of 20 runs; the standard error of the total is at most 0.26).

selector	$\kappa = 200$			$\kappa = 1000$		
	learning	selection	total	learning	selection	total
BBE-noise	4.61	0.22	4.83	13.77	0.95	14.72
BBE-noise, greedy	4.64	0.23	4.87	13.72	0.94	14.65
BBE-ACK	3.08	0.84	3.91	10.08	1.44	11.52
BBE-hybrid	2.90	1.32	4.22	10.07	1.73	11.81
always plain	4.64	0.23	4.87	13.72	0.94	14.65
always interleaved	2.10	13.80	15.89	7.81	8.21	16.02

because the decoder’s gain is far larger for plain codewords.

Acknowledgments reveal early that plain frames lose more codewords than the memoryless prior predicts. BBE-ACK therefore has 19% less regret than BBE-noise for $\kappa = 200$ and 22% less for $\kappa = 1000$, with more selection regret and less learning regret. BBE-hybrid has the lowest learning regret (2.90 against 3.08 for BBE-ACK) and continues to choose the interleaved strategy occasionally in later frames: once B is no longer played, its discounted ACK counts decay and leave the model’s m_0 pseudo-codewords, whose predicted BLER keeps B within reach of the sampler. The selection regret of BBE-noise is 0.22, and almost all of its regret of 4.83 is learning regret. BBE-noise and its greedy version play the same strategy as always sending plain, and the differences among these three rows of Table 3, at most 0.07, are within Monte Carlo error. With full feedback the regret of BBE-noise is 1.72 after 400 frames (Fig. 8a).

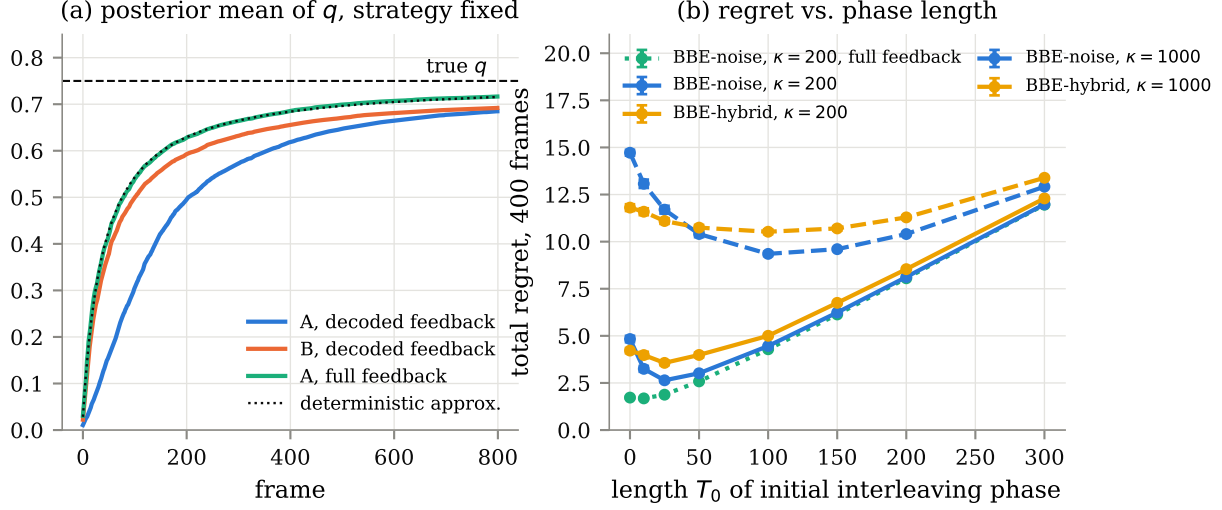


Figure 7: (a) Posterior mean of q with the strategy held fixed: plain frames and interleaved frames with decoded feedback, plain frames with full feedback, and the deterministic approximation. (b) Total regret after 400 frames versus the length of an initial interleaving phase followed by BBE-noise or BBE-hybrid (mean $\pm$ one standard error, 20 runs).

9.5 Initial interleaving phase

Figure 7(a) holds the strategy fixed and follows the posterior mean of q . After 50 frames it is 0.16 when every frame is plain, 0.38 when every frame is interleaved, and 0.41 under full feedback, where the deterministic approximation matches the simulated mean closely. Interleaved frames deliver nearly as much information about q as full feedback.

An initial interleaving phase of T_0 frames uses this directly. Figure 7(b) precedes BBE-noise or BBE-hybrid with such a phase. A phase of $T_0 = 10$ frames, close to the crossing time $\tau = 8$ of Proposition 5, lowers the regret of BBE-noise for $\kappa = 200$ from 4.83 to 3.25. Over a grid of eight lengths the regret is lowest at $T_0 = 25$ for $\kappa = 200$ (2.64) and at $T_0 = 100$ for $\kappa = 1000$ (9.35, against 14.72). The phase helps BBE-hybrid less (minimum 3.56 at $\kappa = 200$), since it interleaves early on its own.

Under full feedback the regret stays between 1.7 and 1.9 for $T_0 \leq 25$ and rises beyond. Interleaved frames carry no extra information there, and the phase helps only by using the cheaper strategy while the decoder is unmatched, so its useful length is close to the crossing time. Under decoded feedback the best phase is about three times longer, in line with the faster growth of the posterior mean of q under interleaving, and it grows with κ , as Proposition 5(iv) predicts for the crossing time.

9.6 Initial knowledge

Table 4 varies the quality of the initial estimate $\hat{\pi}_1$ and the prior strength κ . A log-normal error in $\hat{\pi}_1$ with $\sigma = 0.5$ changed the regret by less than one standard error. About 500 transitions out of state 0 are observed per frame, and they correct p within one or two frames. The prior strength determines which evidence matters most. With $\kappa = 20$ the noise patterns override the memoryless prior within a few frames, and BBE-noise has regret 0.71, lower than BBE-hybrid (1.16). With $\kappa = 200$ and

Table 4: Total regret over 400 frames (mean $\pm$ standard error, 20 runs) with an exact or noisy ($\hat{\pi}_1 = \pi_1 e^{\sigma \xi}$, $\xi \sim \mathcal{N}(0,1)$) initial error rate and three prior strengths.

selector	$\hat{\pi}_1$	prior strength κ		
		20	200	1000
BBE-noise	exact	0.71 ± 0.04	4.83 ± 0.17	14.72 ± 0.16
BBE-noise	noisy ($\sigma = 0.5$)	0.68 ± 0.05	4.81 ± 0.17	14.68 ± 0.18
BBE-hybrid	exact	1.16 ± 0.10	4.22 ± 0.14	11.81 ± 0.18
BBE-hybrid	noisy ($\sigma = 0.5$)	1.11 ± 0.08	4.25 ± 0.15	11.80 ± 0.18

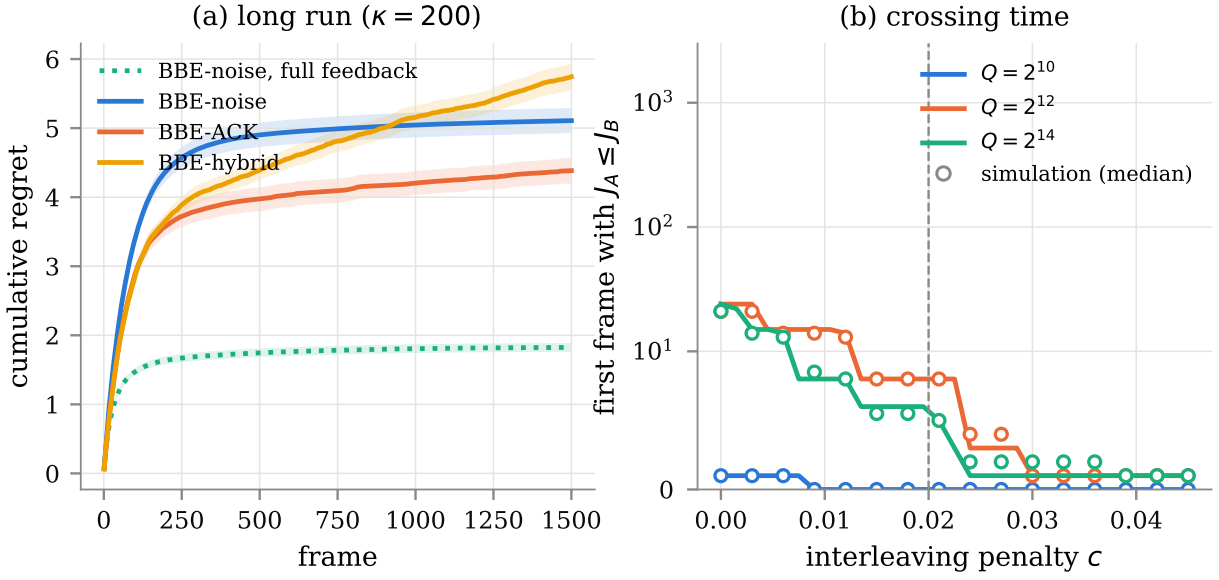


Figure 8: (a) Cumulative regret over 1500 frames ($\kappa = 200$, 20 runs), with decoded feedback and, for BBE-noise, with full feedback. (b) First frame in which the plain strategy is cheaper than the interleaved one, versus the penalty c : deterministic approximation (lines) and median of simulated runs with full feedback (markers). The dashed line marks $c = 0.02$.

$\kappa = 1000$ the acknowledgments carry the early signal, as described above.

9.7 Long horizon and crossing time

Figure 8(a) extends the horizon to 1500 frames. With full feedback, BBE-noise adds 0.10 to its regret after frame 400, in line with Theorem 1. With decoded feedback, BBE-noise goes from 4.83 at frame 400 to 5.11 at frame 1500. BBE-ACK stays below BBE-noise throughout and reaches 4.38. BBE-ACK and BBE-hybrid keep a small positive slope, because discounting makes them return to the interleaved strategy, and BBE-hybrid passes BBE-noise at frame 917. Figure 8(b) compares the crossing time of Proposition 5 with simulated runs under full feedback for three budgets. The two agree, including the steps caused by changes of the query order. The crossing is immediate above c_{mem}^* , which is 0.009, 0.050 and 0.052 for $Q = 2^{10}, 2^{12}, 2^{14}$. On this channel $c_{\text{mat}}^* < 0$ for all three budgets, so the crossing occurs for every $c \geq 0$.

10 Conclusion

This paper introduced Bayesian bandit encoding, a framework in which a Bayesian learner fed by the noise patterns that a GRAND receiver recovers, and by acknowledgments, maintains channel and action posteriors that select the encoding, by Thompson sampling, and order the decoder's guesses. We analyzed BBE on a two-state Markov channel, where the query order and the posterior update have closed forms and the BLER has a closed form under an independent false-hit model. With these we showed that interleaving is asymptotically dominated for a decoder that knows the channel, that the best encoding changes as the decoder learns, and that when every frame reveals its noise the decoder reaches the matched query order after finitely many frames and the number of frames more than δ from optimal is bounded uniformly in time, with explicit constants. With the feedback that GRAND provides, the long-run estimate at the operating point gives a decoder whose BLER matches the matched decoder to four digits. In simulation all BBE variants learn the bursts, and BBE-ACK and BBE-hybrid, whose selection uses acknowledgments, or a short initial interleaving phase, find the switch from interleaved to plain transmission sooner when the receiver starts confident that the channel is memoryless.

There are several possible directions for future work. A regret bound under decoded feedback would build on the bias bound of [Lemma 6](#) and on the speed at which each strategy moves the estimate into the margin of [Assumption 1](#). Selection rules that value the burst statistics an interleaved frame returns, such as information-directed sampling [\[19\]](#), can make the exploration explicit. Finally, the same loop applies to larger action sets, to decoder parameters such as the query budget, and to other channel families such as the multi-interferer channel of [\[5\]](#), where the posterior and the performance model are approximate.

AI Use Statement

The author has used Claude (Anthropic) for writing code and simulations, for help with the analysis and proofs, and for paper writing, editing, proofreading and formatting.

References

- [1] R. Combes, J. Ok, A. Proutière, D. Yun, and Y. Yi, "Optimal rate sampling in 802.11 systems: Theory, design, and implementation," *IEEE Trans. Mobile Comput.*, vol. 18, no. 5, pp. 1145–1158, 2019.
- [2] H. Qi, Z. Hu, X. Wen, and Z. Lu, "Rate adaptation with Thompson sampling in 802.11ac WLAN," *IEEE Commun. Lett.*, vol. 23, no. 10, pp. 1888–1892, 2019.
- [3] W. An, M. Médard, and K. R. Duffy, "Keep the bursts and ditch the interleavers," *IEEE Trans. Commun.*, vol. 70, no. 6, pp. 3655–3667, 2022.
- [4] K. R. Duffy, J. Li, and M. Médard, "Capacity-achieving guessing random additive noise decoding," *IEEE Trans. Inf. Theory*, vol. 65, no. 7, pp. 4023–4040, 2019.
- [5] B. Krishnamachari, "Closed-loop Bayesian bandit encoder with GRAND receiver for a bursty interference channel," unpublished manuscript, arXiv:2607.15404, 2026.
- [6] B. Krishnamachari, "Scaling Bayesian bandit encoding with shared learning," unpublished manuscript, arXiv:2609.06293, 2026.

- [7] A. Solomon, K. R. Duffy, and M. Médard, “Soft maximum likelihood decoding using GRAND,” in *Proc. IEEE ICC*, 2020.
- [8] K. R. Duffy, W. An, and M. Médard, “Ordered reliability bits guessing random additive noise decoding,” *IEEE Trans. Signal Process.*, vol. 70, pp. 4528–4542, 2022.
- [9] H. K. Miyamoto and S. Yang, “Universal decoding over finite-state additive channels via noise guessing,” arXiv:2501.12971, 2025.
- [10] E. Arkan, “An inequality on guessing and its application to sequential decoding,” *IEEE Trans. Inf. Theory*, vol. 42, no. 1, pp. 99–105, 1996.
- [11] D. Malone and W. G. Sullivan, “Guesswork and entropy,” *IEEE Trans. Inf. Theory*, vol. 50, no. 3, pp. 525–526, 2004.
- [12] M. M. Christiansen and K. R. Duffy, “Guesswork, large deviations, and Shannon entropy,” *IEEE Trans. Inf. Theory*, vol. 59, no. 2, pp. 796–802, 2013.
- [13] T. Lattimore and R. Munos, “Bounded regret for finite-armed structured bandits,” in *Proc. NeurIPS*, 2014.
- [14] O. Atan, C. Tekin, and M. van der Schaar, “Global bandits,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 12, pp. 5798–5811, 2018.
- [15] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley, 2006.
- [16] O. Marchal and J. Arbel, “On the sub-Gaussianity of the Beta and Dirichlet distributions,” *Electron. Commun. Probab.*, vol. 22, paper 54, 2017.
- [17] G. Bartók, D. P. Foster, D. Pál, A. Rakhlin, and Cs. Szepesvári, “Partial monitoring—classification, regret bounds, and algorithms,” *Math. Oper. Res.*, vol. 39, no. 4, pp. 967–997, 2014.
- [18] N. Alon, N. Cesa-Bianchi, C. Gentile, and Y. Mansour, “From bandits to experts: A tale of domination and independence,” in *Proc. NeurIPS*, 2013.
- [19] D. Russo and B. Van Roy, “Learning to optimize via information-directed sampling,” *Oper. Res.*, vol. 66, no. 1, pp. 230–252, 2018.
- [20] T. L. Lai and H. Robbins, “Asymptotically efficient adaptive allocation rules,” *Adv. Appl. Math.*, vol. 6, no. 1, pp. 4–22, 1985.